

PWS Detailed Overview

The Problem: The Traditional Examination Process

Traditionally, practical and internal exams are conducted on paper. College teachers manually create the question papers, and the colleges then print copies for every student.

Traditional Practical Exams

For practical exams, there is a standard procedure where the teacher places all the question slips face down on a table so students cannot see the questions. Students then pick a random slip, which becomes their specific question paper. Each student receives two documents: the question slip and an answer sheet.

The student's role is to read the question, write the code manually on the answer sheet, and then type that exact same code into a PC or laptop. After typing the code, the teacher goes to the student's computer, checks the code, and gives approval for submission. To give approval, the teacher must physically sign the student's slip, allowing the student to submit both the question and answer papers.

Next, the teacher manually checks the answers and assigns marks. The papers are then sent to a moderator for a second review. To digitize this paperwork, the teacher must manually enter the data into an Excel sheet—creating columns for the student's name, roll number, attempted questions, and assigned marks. Additionally, while the exam is ongoing, the teacher must manage a printed attendance sheet that students have to sign to mark themselves present.

Traditional Internal Exams (MCQs)

For internal exams, the teacher creates a question bank of MCQs and manually selects questions to design the paper, similar to the practical exams. This paper is sent to a moderator to check for typos or errors. Once approved, it is printed.

Students receive a question paper and an answer sheet. They read the MCQs and write the correct answers on the answer sheet. The teacher collects all the papers, ensures the attendance sheet is signed, and carries the physical stack of papers back to the staffroom. Sitting at their desk, the teacher manually grades each paper and assigns marks. A moderator then re-checks the papers to ensure the grading is correct. After receiving the moderator's approval, the teacher digitizes the work by manually entering the student names, roll numbers, attempted questions, and marks into an Excel sheet to generate results and determine who passed or failed.

The Drawbacks

This traditional method relies entirely on paper and manual labor, creating a chaotic and hectic process. It is highly unreliable because physical papers can easily get wet or torn. Furthermore, it is impractical to store these papers long-term, as they can easily get misplaced over the years.

The Solution: PRACTICAL WORKFLOW SYSTEM

To solve these issues, we have developed the PRACTICAL WORKFLOW SYSTEM, an end-to-end software solution for teachers, students, and college administrators.

For College Admins, the platform provides a hassle-free way to manage all teachers and generate results instantly. After an exam ends, admins have access to a real-time dashboard where they can check every student's scores within seconds.

Teachers are provided with an online platform where they can simply log in, create an exam, and launch it within minutes—turning hours of tedious work into a quick, frictionless process.

Digital Practical Exams

For practical exams, the teacher simply uploads their question bank into the system. The software automatically analyzes the questions and randomly distributes them to students. This replicates the traditional randomness of picking paper slips, but in a fully digital format. Each student receives a randomly assigned digital slip.

The exams are conducted in a locked digital environment to prevent cheating. Because the screen is locked, students cannot access ChatGPT, Gemini, or any other external applications, ensuring a completely fair exam.

Students no longer have to do the redundant work of writing code on paper and then typing it on a device. They simply write the code once directly into the system, saving time and effort. After writing the code, students upload their output results to the system.

The teacher does not even need to leave their chair. They have access to a live dashboard displaying real-time data. When students upload their answers, they request approval through the system. The teacher can view the student's code and attached output directly on their screen and approve it from their seat. Furthermore, attendance is fully automated—as soon as a student logs into the exam, they are marked present.

Grading is also done online, which makes it much simpler for the teacher compared to shuffling through physical papers. The teacher views the question and the student's answer side-by-side on the screen to assign marks. Once grading is complete, the teacher can instantly download the attendance sheet and the final results. Everything, from exam creation to result generation, is fully automated.

Advanced Features for Practical Exams

- **Slip Changing & Custom Assignment:** If a student cannot answer a randomly assigned question, the teacher can change their slip or manually assign a specific question to them.
- **Reusable Templates:** If a teacher conducts similar exams repeatedly, they can create a single template to use multiple times.
- **Template Sharing:** A teacher can share their created templates with other teachers, saving time across the faculty.

- **Live Session Sharing & Remote Monitoring:** Teachers can share live exam sessions. For example, Teacher A can share their ongoing session with Teacher B, allowing Teacher B to act as a moderator and monitor the exam from anywhere.
- **Exam Resumption:** If a teacher accidentally ends an exam, they can easily resume it from exactly where it stopped.
- **Student Buffer:** If a student was missing from the initial Excel list, the teacher can manually add them to a live, ongoing exam.

Digital Internal Exams (MCQs)

For internal exams, the teacher only needs to upload their MCQ question bank and the student list. The software automatically analyzes the MCQs and randomly picks questions to fulfill the total marks specified by the teacher.

The questions are automatically distributed to the students' screens upon logging in. Once students submit their exams, the software automatically evaluates the correct and incorrect answers and assigns marks instantly. The teacher can then easily review the answered questions.

From creating the exam to grading and generating results, the entire internal exam process is automated. This software transforms a manual, chaotic process into a 100% paperless, digitalized, peaceful, and fair examination system.

Security Measures Implemented

- **Authentication & Authorization:** Strict Role-Based Access Control (RBAC), Firebase ID Token verification for backend endpoints, and multi-tenant isolation ensuring users can only access their own college's data.
- **Database Security (Firestore Rules):** Highly scoped access rules enforce that users (especially students) can only read and write to their specific documents. Critical actions enforce state-machine integrity and server-side subscription validation.
- **Storage Security:** Prevention of path traversal vulnerabilities, strict MIME-type and size limits on uploads, and restricted read/write ownership.
- **Network Security:** Strict CORS policies, aggressive rate-limiting on sensitive endpoints (OTP, heartbeats, emails) to prevent DDoS and brute-forcing attacks, and HTML sanitization for email templates.
- **Anti-Cheating & Auditing:** A tamper-proof server-side audit logging system, a "soft delete" architecture to prevent permanent data loss, atomic transactions for critical actions, and a server-enforced monotonic heartbeat to prevent students from bypassing tab-switch tracking.